

Calculating averages by hand.

horsebean chicken weight average

```
In [1]: (179+160+136+227+217+168+108+124+143+140) / 10
```

```
Out[1]: 160.2
```

linseed chicken weight average

```
In [2]: (309+229+181+141+260+203+148+169+213+257+244+271) / 12
```

```
Out[2]: 218.75
```

soybean chicken weight average

```
In [3]: (243+230+248+327+329+250+193+271+316+267+199+171+158+248)
```

```
Out[3]: 246.42857142857142
```

sunflower chicken weight average

```
In [4]: (423+340+392+339+341+226+320+295+334+322+297+318) / 12
```

```
Out[4]: 328.9166666666667
```

meatmeal chicken weight average

```
In [5]: (325+257+303+315+380+153+263+242+206+344+258) / 11
```

```
Out[5]: 276.90909090909093
```

casein chicken weight average

```
In [6]: (368+390+379+260+404+318+352+359+216+222+283+332) / 12
```

```
Out[6]: 323.5833333333333
```

Putting our averages in a table for a prediction.

```
In [7]: import pandas as pd
```

```
In [8]: pd.DataFrame({"average weight": [160.2, 218.75, 246.42, 328.91, 276.90, 323.58],
```

```
Out[8]:
```

	average weight	feed
0	160.20	horsebean
1	218.75	linseed
2	246.42	soybean
3	328.91	sunflower
4	276.90	meatmeal
5	323.58	casein

Printing our data for each type of feed.

```
In [9]: horsebean_data = [179,160,136,227,217,168,108,124,143,140]  
linseed_data = [309,229,181,141,260,203,148,169,213,257,24  
soybean_data = [243,230,248,327,329,250,193,271,316,267,19  
sunflower_data = [423,340,392,339,341,226,320,295,334,322,  
meatmeal_data = [325,257,303,315,380,153,263,242,206,344,2  
casein_data = [368,390,379,260,404,318,352,359,216,222,283
```

```
In [10]: import numpy as np
```

```
In [11]: horsebean_max = np.max(horsebean_data)  
horsebean_min = np.min(horsebean_data)  
horsebean_mean = np.mean(horsebean_data)  
horsebean_variance = np.var(horsebean_data)  
horsebean_stddev = np.std(horsebean_data)  
horsebean_median = np.median(horsebean_data)  
  
print("horsebean max:{0:d}".format(horsebean_max))  
print("horsebean min:{0:d}".format(horsebean_min))
```

```

print("horsebean mean:{0:8.4f}".format(horsebean_mean))
print("horsebean variance:{0:8.4f}".format(horsebean_variance))
print("horsebean std:{0:8.4f}".format(horsebean_standarddev))
print("horsebean median:{0:8.4f}".format(horsebean_median))

```

```

horsebean max:227
horsebean min:108
horsebean mean:160.2000
horsebean variance:1342.7600
horsebean std: 36.6437
horsebean median:151.5000

```

```

In [12]: linseed_max = np.max(linseed_data)
linseed_min = np.min(linseed_data)
linseed_mean = np.mean(linseed_data)
linseed_variance = np.var(linseed_data)
linseed_standarddev = np.std(linseed_data)
linseed_median = np.median(linseed_data)

print("linseed max:{0:d}".format(linseed_max))
print("linseed min:{0:d}".format(linseed_min))
print("linseed mean:{0:8.4f}".format(linseed_mean))
print("linseed variance:{0:8.4f}".format(linseed_variance))
print("linseed std:{0:8.4f}".format(linseed_standarddev))
print("linseed median:{0:8.4f}".format(linseed_median))

```

```

linseed max:309
linseed min:141
linseed mean:218.7500
linseed variance:2501.1875
linseed std: 50.0119
linseed median:221.0000

```

```

In [13]: soybean_max = np.max(soybean_data)
soybean_min = np.min(soybean_data)
soybean_mean = np.mean(soybean_data)
soybean_variance = np.var(soybean_data)
soybean_standarddev = np.std(soybean_data)
soybean_median = np.median(soybean_data)

print("soybean max:{0:d}".format(soybean_max))
print("soybean min:{0:d}".format(soybean_min))
print("soybean mean:{0:8.4f}".format(soybean_mean))
print("soybean variance:{0:8.4f}".format(soybean_variance))
print("soybean std:{0:8.4f}".format(soybean_standarddev))
print("soybean median:{0:8.4f}".format(soybean_median))

```

```
soybean max:329
soybean min:158
soybean mean:246.4286
soybean variance:2720.6735
soybean std: 52.1601
soybean median:248.0000
```

```
In [14]: sunflower_max = np.max(sunflower_data)
sunflower_min = np.min(sunflower_data)
sunflower_mean = np.mean(sunflower_data)
sunflower_variance = np.var(sunflower_data)
sunflower_stddev = np.std(sunflower_data)
sunflower_median = np.median(sunflower_data)

print("sunflower max:{0:d}".format(sunflower_max))
print("sunflower min:{0:d}".format(sunflower_min))
print("sunflower mean:{0:8.4f}".format(sunflower_mean))
print("sunflower variance:{0:8.4f}".format(sunflower_variance))
print("sunflower std:{0:8.4f}".format(sunflower_stddev))
print("sunflower median:{0:8.4f}".format(sunflower_median))
```

```
sunflower max:423
sunflower min:226
sunflower mean:328.9167
sunflower variance:2186.2431
sunflower std: 46.7573
sunflower median:328.0000
```

```
In [15]: meatmeal_max = np.max(meatmeal_data)
meatmeal_min = np.min(meatmeal_data)
meatmeal_mean = np.mean(meatmeal_data)
meatmeal_variance = np.var(meatmeal_data)
meatmeal_stddev = np.std(meatmeal_data)
meatmeal_median = np.median(meatmeal_data)

print("meatmeal max:{0:d}".format(meatmeal_max))
print("meatmeal min:{0:d}".format(meatmeal_min))
print("meatmeal mean:{0:8.4f}".format(meatmeal_mean))
print("meatmeal variance:{0:8.4f}".format(meatmeal_variance))
print("meatmeal std:{0:8.4f}".format(meatmeal_stddev))
print("meatmeal median:{0:8.4f}".format(meatmeal_median))
```

```
meatmeal max:380
meatmeal min:153
meatmeal mean:276.9091
meatmeal variance:3829.1736
meatmeal std: 61.8803
meatmeal median:263.0000
```

```
In [16]: casein_max = np.max(casein_data)
casein_min = np.min(casein_data)
casein_mean = np.mean(casein_data)
casein_variance = np.var(casein_data)
casein_stddev = np.std(casein_data)
casein_median = np.median(casein_data)

print("casein max:{0:d}".format(casein_max))
print("casein min:{0:d}".format(casein_min))
print("casein mean:{0:8.4f}".format(casein_mean))
print("casein variance:{0:8.4f}".format(casein_variance))
print("casein std:{0:8.4f}".format(casein_stddev))
print("casein median:{0:8.4f}".format(casein_median))

casein max:404
casein min:216
casein mean:323.5833
casein variance:3805.7431
casein std: 61.6907
casein median:342.0000
```

Calculating the averages of chicken weight of chickens that were fed a different type of chicken feed, reveals that **sunflower** makes the heaviest chicken at 328.91 grams. The second heaviest chicken was fed **casein** and has an average of 323.58 grams. **Meatmeal** was the third heaviest average weight of the chicken at at 276.90 grams. **Soybean** was next heaviest after meatmeal, **linseed** was after soybean, and **horsebean** was the least heavy average weight of all other chicken feed. Other than the average or mean, other factors can be involved when finding the most effective feed.

Variance and standard deviation show how values are spread across this distribution. Although horsebean has the lowest weight, the variance is also the lowest, at 1342.75, the second lowest variance was

sunflower at 2186.24. The highest variance was meatmeal at 3829.17, and the second highest was casein at 3805.74.

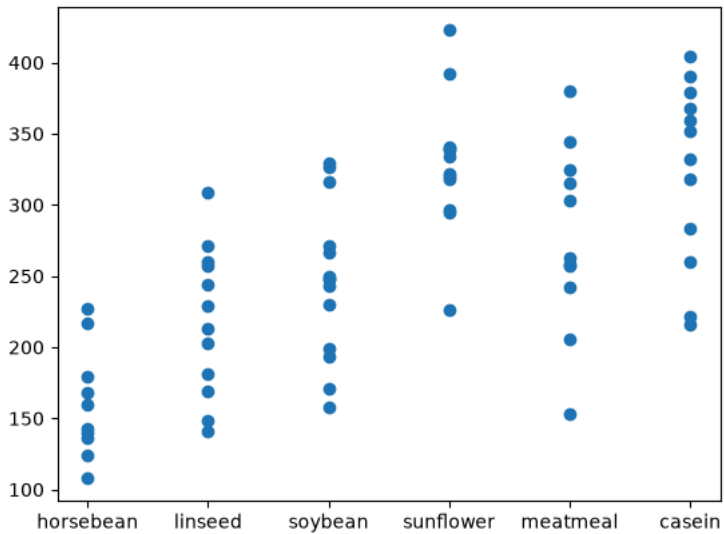
```
In [17]: print("Variance")
print("-"*15)
print("horsebean ", horsebean_variance)
print("linseed  ", linseed_variance)
print("soybean  ", soybean_variance)
print("sunflower ", sunflower_variance)
print("meatmeal ", meatmeal_variance)
print("casein  ", casein_variance)
```

```
Variance
-----
horsebean  1342.76
linseed    2501.1875
soybean    2720.673469387755
sunflower  2186.2430555555557
meatmeal   3829.173553719009
casein     3805.743055555556
```

Importing our csv file and plotting the dataset.

```
In [18]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv('chickwts.csv')
plt.scatter(df.feed, df.weight)
```

```
Out[18]: <matplotlib.collections.PathCollection at 0x1a918b14e30>
```



From the scatter plot above we can see that horsebean feed is the least spread out, meaning that the chickens are predictable but too light. Sunflower seed feed, is the most high and reveals to be the heaviest chicken producer. This could be the best feed because the lightest chicken from the sunflower feed (min) is actually very close to the heaviest chicken from horsebean feed (max). Horsebean max is 227, and sunflower min is 226.

The median tells us the middle value of the range of data. The highest is **sunflower** seed for chickens at 328 grams and the next highest weight is **casein** at 342 grams. I would predict that **sunflower** is the best feed and **casein** is the next best feed. Sunflower wins as the best middle performer and the heaviest chicken weight. (Shah | "Intro to DS using Py." | Code 5.1 - 5.13)

In []: